

DevOps and Automation - Lab 1

Task 3: Automate Application Deployment Jenkins CI/CD Deployment Automation Report

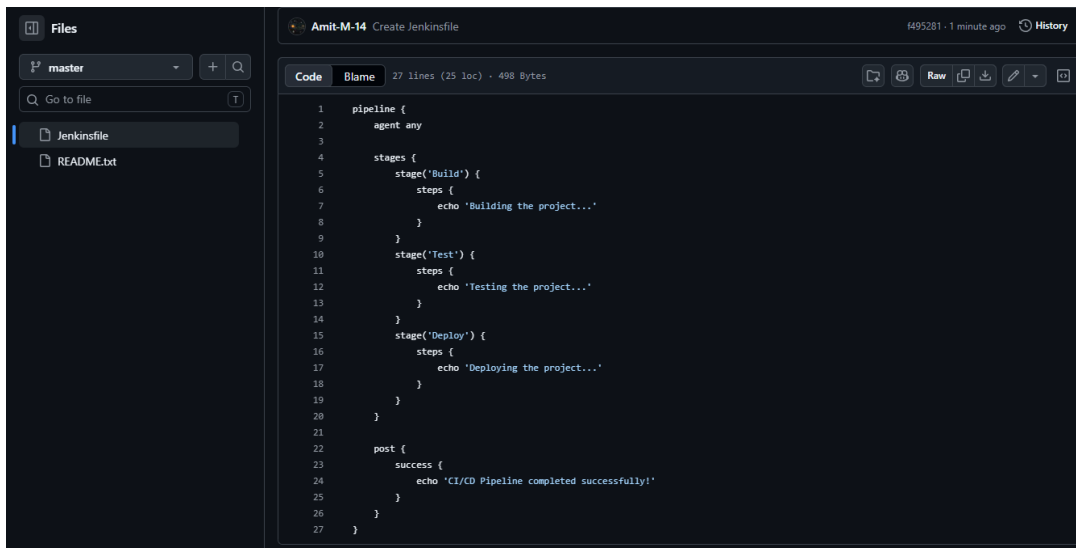
Student	Amit Mohanty
Repository	devops-git-lab
Jenkins Job	devops-git-lab-pipeline
Branch	master
SCM	Git / GitHub

1. Objective

To automate application deployment using Jenkins by defining Build, Test and Deploy stages, executing the pipeline from source control, and verifying the successful deployment result.

2. Jenkinsfile

The following screenshot shows the Jenkinsfile stored in the GitHub repository. It defines a declarative pipeline with Build, Test and Deploy stages and a post-success action.



```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Build') {
6       steps {
7         echo 'Building the project...'
8       }
9     }
10    stage('Test') {
11      steps {
12        echo 'Testing the project...'
13      }
14    }
15    stage('Deploy') {
16      steps {
17        echo 'Deploying the project...'
18      }
19    }
20  }
21
22  post {
23    success {
24      echo 'CI/CD Pipeline completed successfully!'
25    }
26  }
27 }
```

Figure 1. Jenkinsfile defining the Build, Test and Deploy pipeline stages.

3. Pipeline Workflow

Build	Jenkins executes the build step and displays the building message.
Test	Jenkins executes the testing step and displays the testing message.
Deploy	Jenkins executes the deployment step and confirms successful application deployment.
Post - Success	Jenkins prints the CI/CD pipeline completion message after a successful run.

4. Jenkins Execution Evidence

The Jenkins execution completed successfully. The console output shows the Build, Test and Deploy stages, the successful post action, and the final status Finished: SUCCESS.

```
Testing the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Figure 2. Jenkins Console Output showing Build, Test, Deploy and Finished: SUCCESS.

The screenshots below are from the Jenkins/GitHub work completed for this lab.

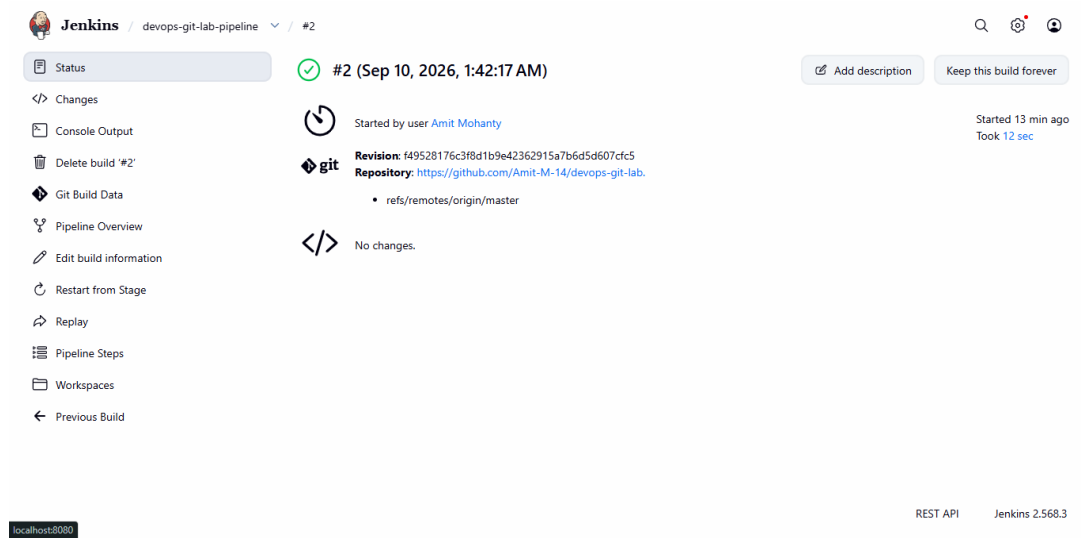


Figure 3. Jenkins Build #2 completed successfully. The page shows the repository, master branch revision, and successful build status.

```
Started by user Amit Mohanty
Obtained Jenkinsfile from git https://github.com/Amit-M-14/devops-git-lab.
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline\.git #
timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Amit-M-14/devops-git-lab. # timeout=10
Fetching upstream changes from https://github.com/Amit-M-14/devops-git-lab.
> git.exe --version # timeout=10
> git --version # 'git version 2.53.0.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/Amit-M-14/devops-git-lab.
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision f49528176c3f8d1b9e42362915a7b6d5d607cfc5 (refs/remotes/origin/master)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f f49528176c3f8d1b9e42362915a7b6d5d607cfc5 # timeout=10
Commit message: "Create Jenkinsfile"
```

Figure 4. Jenkins Console Output showing Git checkout from the GitHub repository and execution of the pipeline.

5. Observation

The Jenkins pipeline successfully executes the defined stages in sequence. The Jenkinsfile provides the pipeline structure, while the Jenkins console confirms that the deployment stage and the complete CI/CD pipeline finished successfully.